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Auto Insurance Pure Premium Calculation

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My name is Jarrett Baker. I am a senior studying actuarial science, and I wanted to get some more experience within the actuarial field. In December 2025, I set out to create a project that would be very applicable to an actuarial job, while also using a dataset that is much larger than any other project I have used previously. I wanted to figure out the pricing for an auto insurance company, as I am familiar with some of the policies and procedures that would come from a property and casualty style dataset.

My goals with this project are the following: to get more familiar with actuarial modeling, being able and more comfortable to work with a bigger dataset, being able to build a model that can encapsulate the amount of claims along with the severity of them too. As well, being able to work with an open mind, as I am fairly new to this, and being able to work very well with criticism and figuring out why we use certain tests and what the assumptions of those tests are.

The initial dataset I found on the website Kaggle, and after some further research, I found the data came from the Casualty Actuarial Society (CAS) website, and it came from an unnamed insurance company with French policyholders that contains two separate files. The first is titled freMTPL2freq, where contained are a list of variables contained below with their description directly from the CAS, and the second is freMTPL2sev, where the variables included are the claimID and the severity of the claims. An important note that needs to be mentioned is that some of the claim amounts are fixed based on the French IRSA-IDA claim convention (Convention – IRSA), which states if the damage is between €2030 and €6500, the recourse is a flat rate within that range. There is also 678,013 rows in this dataset, with 95% of those rows having zero claims

freMTPL2freq contains 11 columns:

IDpol The policy ID (used to link with the claims dataset)

ClaimNb Number of claims during the exposure period.

Exposure The period of exposure for a policy, in years.

VehPower The power of the car (ordered values).

VehAge The vehicle age, in years.

DrivAge The driver age, in years (in France, people can drive a car at 18).

BonusMalus Bonus/malus, between 50 and 350: 100 means malus in France.

VehBrand The car brand (unknown categories).

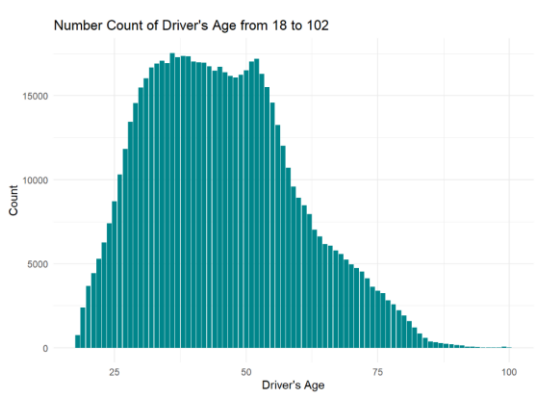
VehGas The car gas, Diesel or regular.

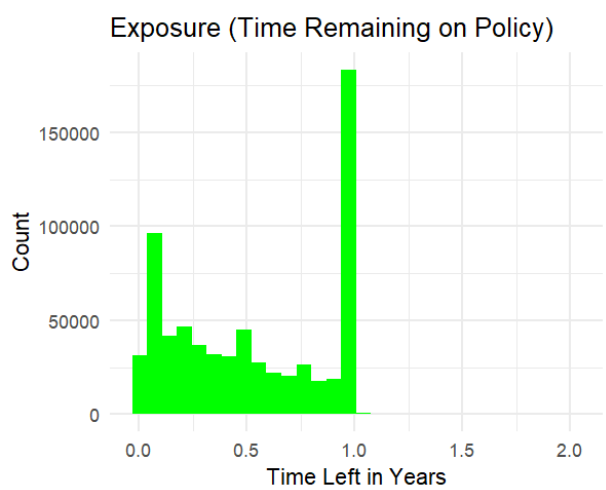
Area The density value of the city community where the car driver lives in: from "A" for rural area to "F" for urban centre.

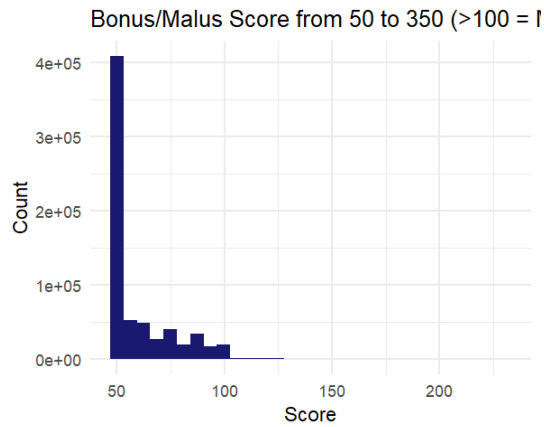
Density The density of inhabitants (number of inhabitants per square-kilometer) of the city where the car driver lives in.

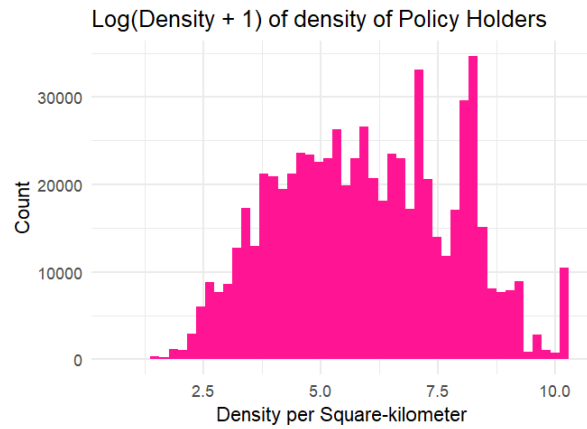
Region The policy region in France (based on the 1970-2015 classification).

Modeling each variable, we can see some important information about some of them. I have included the graphs below of the ones I am discussing. The main age of the drivers on the policies are between the age of 25 to 55 having very similar amounts, then the tails slowly decrease out from those. The exposure the company has, is fairly even throughout the year, other than the ones that have almost a year left on their policy. This is good to make a comment that it could either be the beginning of the year or a popular time to renew their insurance, or they might have had a deal or a huge flood of customers wanting to buy insurance. The bonus/malus score, meaning a credit score for the policyholders, if they are under the 100 score, they receive a slightly better price because of their risk to have a claim in the future. If it is above 100, that is the riskier policyholders and may have to charge them a slightly higher rate because of the history of them having more claims. I took the natural logarithm of the density variable because it is much easier to look at the natural logarithm of the density, and the variable is much easier to use in our models.









For this project, I did not cap the premiums or how much the insurance company would pay out in a claim. While this may be unrealistic, I believed this would be easier for me to model and predict, which may not have been true and made this slightly more difficult.

After cleaning the data, the next step was to compare the claims to a model, and see which model fits the closest to it. I chose four different models to test. Poisson, Negative Binomial, Zero Inflation Poisson, and Zero Inflation Negative Binomial. In all of these tests, we are trying to predict the claim number for policyholders, while using the VehPower, VehAge, DrivAge, BonusMalus, Density, Region, and the offset of the Exposure variable. For the normal Poisson, significant overdispersion (taking variance and dividing mean) was present, with a ratio of 2.7186, which is highly above 2. Performing negative binomial tests onto the dataset resulted in a better result than Poisson, based off the lower AIC.

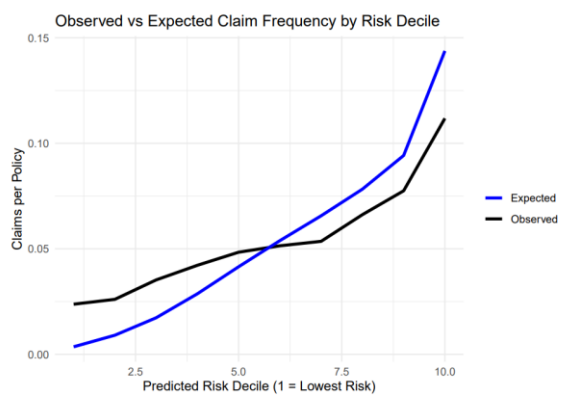
The idea for zero inflation models came from thinking about how many zero claims there are in auto insurance each year. For our zero inflation models, we must slightly change the models, where we divide the variables into two groups: normal Poisson/Negative Binomial and the logistic model trying to see if the zero is a structural zero. All our variables are used in the Poisson/Negative Binomial test, and then for the structural zeros, I used DrivAge and Bonus Malus to see if the groups can help show structural zeros. Creating the following table, I was able to compare the different metrics we would like to compare between these tests.

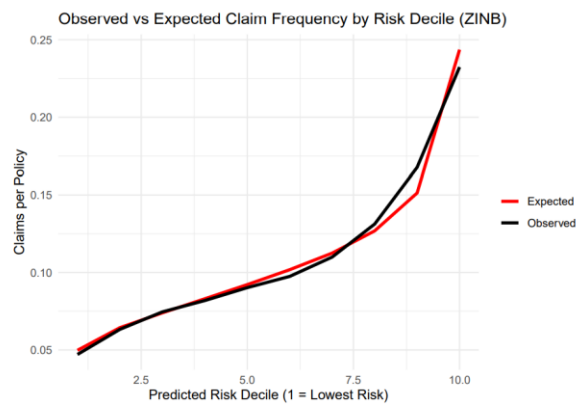
```
comparison_wide
```

```
## # A tibble: 9 x 5
##   Metric          Poisson   NegBin      ZIP     ZINB
##   <chr>             <dbl>    <dbl>    <dbl>    <dbl>
## 1 N               678013   678013   678013   678013
## 2 DF_model            36       36       49       49
## 3 DF_resid        677977   677977   677964   677963
## 4 AIC             286836.  286016.  285809.  285571.
## 5 LogLik         -143382. -142971. -142855. -142735.
## 6 Deviance        217441.  190618.  217101.  217004.
## 7 Pearson_ChiSq  1841084. 1819324. 1771436. 1762631.
## 8 Predicted_Zero  643482.  643084.  644278.  644392.
## 9 Actual_Zero     643953   643953   643953   643953
```

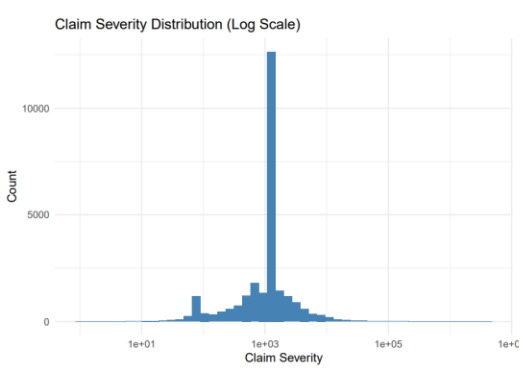
A lot of things were shown through this chart, and I will point out some of the main ones I noticed. Zero Inflation Poisson (ZIP) and Zero Inflation Negative Binomial (ZINB) beat out their non-zero inflation tests in almost every test. Their log likelihoods were higher, with ZINB slightly beating out ZIP. The pearson chi-squared test you are looking for a lower value, and again the ZINB slightly beat out ZIP. But for calculating the closest amount of the predicted zeroes to the actual zeroes, both the zero inflation models beat out the normal ones, but ZIP slightly beat the ZINB here.

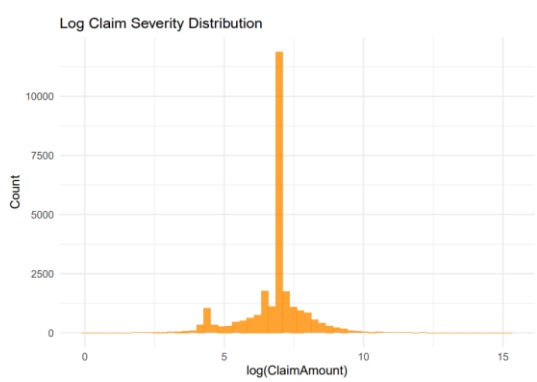
So which model is best for predicting the number of claims? Modeling both types of zero inflation models should help show a better picture of which one to choose, showing the expected vs observed claims per policy when looking at the different predicted risk decile. The Zero Inflation Poisson is modeled in blue, and the Zero Inflation Negative Binomial has the red line. We can see that the expected line is much closer to the observed line in the ZINB case. We will continue to use zero inflation negative binomial forward.



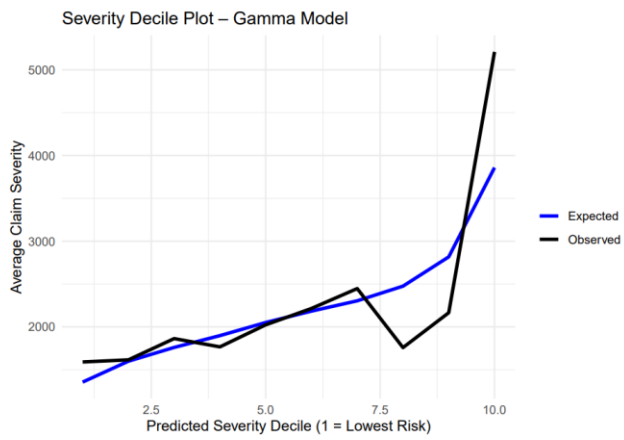


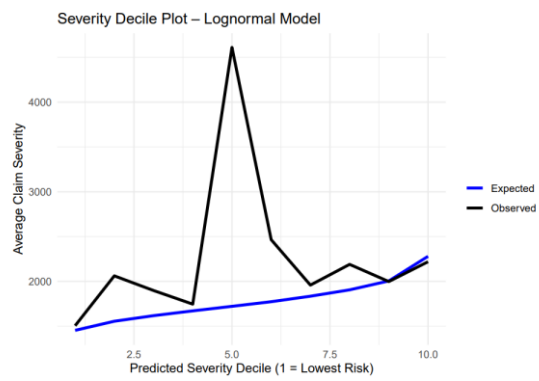
Switching over to looking at the severity part of the dataset, we first can look at the distribution of the natural logarithm of the severities to see its distribution. There are 26,444 claims in the dataset, and the natural log has a massive bar where the median is at €1172. We can see there is a huge amount of claims near that point (could possibly be due to the IRSA-IDA claim convention).





Testing the two main distributions that could help model severity (gamma and lognormal), we can test if either of the two are good predictors or if we would need another test. Looking at both severity plots, we can note some things. The gamma model accurately predicts the lower severity deciles (which most of the claims are at), but has a bit more trouble predicting the tail end of the severity (where it is harder to predict and there are much fewer claims). While the lognormal plot does not look quite as good, we can not that it does a good job predicting that tail end that the gamma has a bit of a trouble with. But we will proceed further with the gamma model, as it does a good job predicting most of the claims compared to the lognormal model. Our data also has a rather large tail, which could be a reason why the lognormal model struggles so much, as well why the gamma model has trouble with that high decile side. The quantiles are listed below to show how much of a rather large tail it has.






```
quantile(severity_model_data$ClaimAmount, probs = c(0.9, 0.95, 0.99, 0.999))
```

```
##        90%        95%        99%      99.9%
##   2767.604   4765.093  16451.224 152223.241
```

Calculating the pure premium (amount of claims * predicted amount of each claim), we will just combine the two models we came up with. Our zero-inflation negative binomial combined with our gamma model, we can predict how much each customer would pay for a pure premium of pure risk for everyone in this dataset. While that may be a little hard to show what each one of our claims are, we can get a couple metrics to see how well our model performs. Our root mean squared error (RMSE) is 3,512, and our mean absolute error is 188. Which means per policy, we are off by $188 per policy if we charge purely premium. Keeping in mind that what was modeled had no maximum where the insurance company would not pay, this can be reasonable (our max was €4,000,000!). Our RMSE is slightly higher than what I would like to see, but there are 95% zeros in the dataset, and our claims contain a couple massive ones.

Through this project, I have learned so much. Before I did not know anything about zero inflation models, and now I am familiar with some of the assumptions and how to model both the

Poisson and negative binomial versions. While there are plenty of things I would do differently next time if I were to redo this, I am very happy with how this project went. Improvements I would make for next time would be to slow down and make sure that I know all of the assumptions of models, when they work well, when they do not work well. I am still unsure if to use zero inflation negative binomial or zero inflation Poisson for the number of claims, but I still stand by the zero inflation negative binomial models. I would certainly put in a cap for the claims, and maybe at the same time, put in a deductible the customer puts in first before the insurance company pays. I think it would be good to see how they each change the models by themselves, and how they change them when both are in the same model together.

Overall, I would say this project is a huge success. I set out and established some goals to do in this project, and I was able to hit almost all of them. I certainly will do another project like this, but I am unsure if I will do it before or after I graduate, when I have more time.

Works Cited

Sarpal, Karan. "FREMTPL - French Motor TPL Insurance Claims Data." *Kaggle*, 7 Nov. 2020, www.kaggle.com/datasets/karansarpal/fremtpl-french-motor-tpl-insurance-claims.

"Convention IRSA - Ida." *Index Assurance*, www.index-assurance.fr/pratique/sinistre/convention-irsa. Accessed 14 January 2026.